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Country Energy Report

July 2026

Malaysia



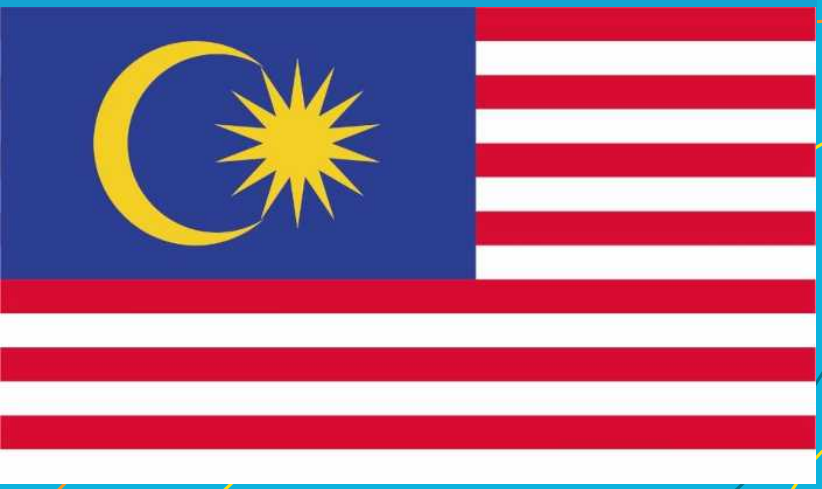


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Overview



Map sources: UNC, Natural Earth. The boundaries and names shown and the designations used on this map do not imply official endorsement or acceptance by the United Nations. Map created in Sep 2013

Highlights

- The GHG emissions intensity is planned to be reduced by 45% in 2030.

  For renewables, the country targets a 40% share in the power capacity mix in 2035 and a 17% share in the energy mix in 2040.

- Two public companies dominate the energy sector: TNB for electricity and Petronas for oil and gas.

- The share of renewables in the power mix has increased by 14 points since 2010 to 20% in 2025. Thermal generation is shared between coal (46%) and gas (32%).

- Final energy consumption has been increasing by 4.5%/year since 2022.

  Diesel prices increased by 25% following the removal of subsidies. Gasoline and diesel are much cheaper than in the most developed Asian countries.

- The National Energy Plan 2022-2040 aims to reduce the share of coal in the power capacity mix from 31% in 2018 to 18.6% in 2040.

  Petronas is partnering with various companies to explore the potential and viability of CCS technology, making Malaysia a key CCS hub in the ASEAN region.

-45%

reduction in GHG intensity by 2030

-50%

coal power plants by 2035

17%

share of renewables in the energy mix in 2040

Table 1 : Economic Indicators

		1990	2000	2010	2019	2022	2023	2024	2025
Population	*million*	17.8	23.0	28.7	33.4	34.7	35.1	35.6	36.0
GDP growth rate	*%/year*	9.0	8.9	7.4	4.4	8.9	3.6	5.1	5.2
GDP/capita	*USD*	2 469	4 084	8 899	10 920	11 748	11 379	11 867	13 118
Inflation rate	*%/year*	2.6	1.5	1.6	0.66	3.4	2.5	1.8	1.4
Exchange rate	*lc/USD*	2.7	3.8	3.2	4.1	4.4	4.6	4.6	4.3

Energy security		1990	2000	2010	2019	2022	2023	2024	2025
Energy independence rate	*%*	100	100	100	100	96	94	91	90
Share of oil imported(+) exported(-)	*%*	-138	-44	-13	7	16	27	32	30

Energy efficiency		1990	2000	2010	2019	2022	2023	2024	2025
Total consumption/GDP *	*koe/USD15p*	0.094	0.127	0.121	0.103	0.106	0.103	0.102	0.100
Total consumption/GDP *	*2015=100*	87.3	118	112	95.7	97.9	95.3	95.0	92.4
Rate of T&D power losses	*%*	7.9	8.2	7.1	7.2	7.2	7.2	7.4	6.2
Efficiency of thermal power plants	*%*	38.5	39.7	31.4	37.4	37.7	37.9	38.4	38.2

CO_2 emissions		1990	2000	2010	2019	2022	2023	2024	2025
CO_2 emissions/GDP *	*kCO_2/USD15p*	0.261	0.292	0.299	0.250	0.257	0.248	0.246	0.243
CO_2 emissions/capita	*tCO_2/cap.*	2.8	4.8	6.2	6.9	7.3	7.2	7.4	7.6

** at purchasing power parity*

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Energy Use & Price by Sector



Final energy consumption has been increasing by 4.5%/year since 2022.

Diesel price increased by 25% following the removal of subsidies. Gasoline and diesel are much cheaper than in the most developed Asian countries.

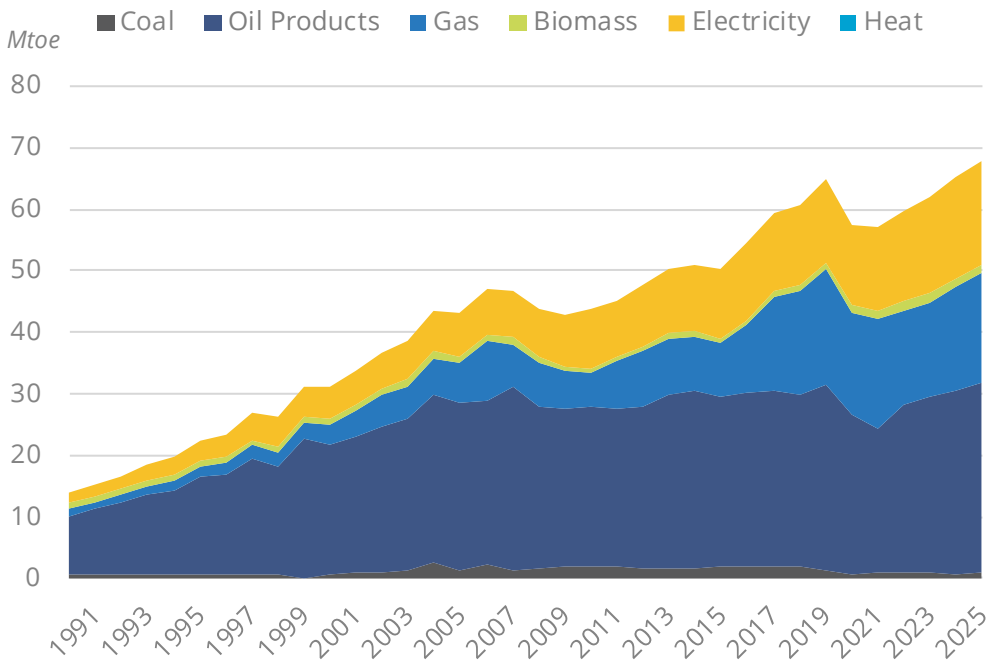
FINAL ENERGY DEMAND

Final energy consumption has been increasing by an average of 4.5%/year since 2022, a pace similar to that observed over 2010–2019.

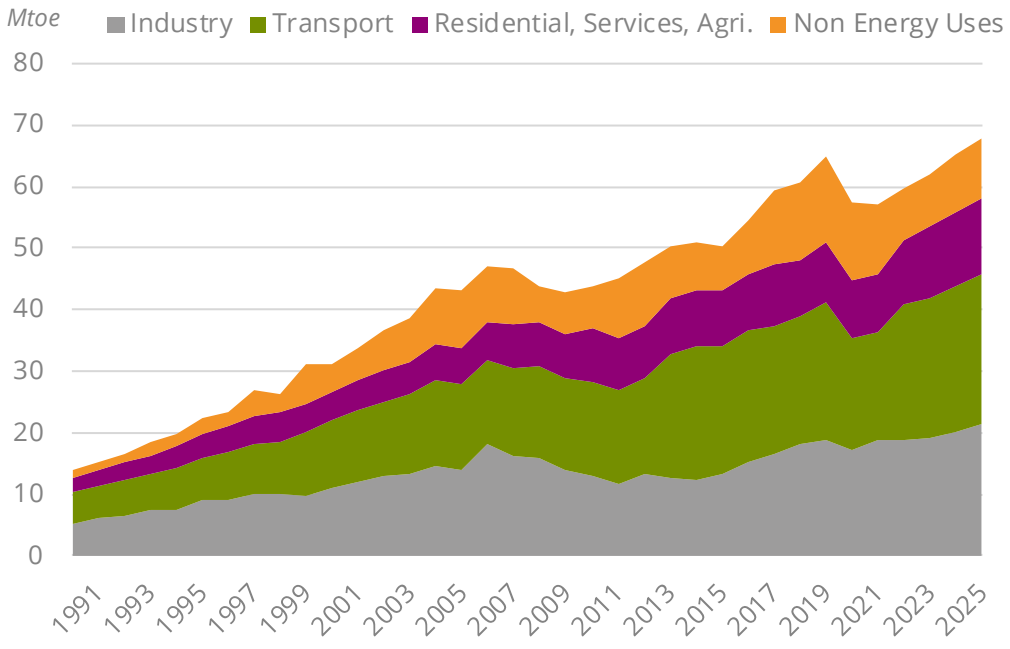
Oil products account for almost half of total demand (46% in 2025), followed by gas (26%) and electricity (24%). The share of oil has fallen by 14 percentage points since 2010, while gas has gradually taken its place, rising by 13 pp. In our EnerBlue scenario, the share of electricity is expected to reach 30% by 2030 (compared with 26% in non-OECD countries) and 34% by 2050 (37% for non-OECD).

The transport sector represents more than one-third of total final consumption (36% in 2025), followed by industry (31%), the residential and services sectors (18%), and non-energy uses (15%). These shares have remained broadly stable since 2010.

Graph 16: Final Demand Trends by Energy Source



Graph 17: Final Consumption by Sector



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Issues & Prospects



The National Energy Plan 2022-2040 aims to reduce the share of coal in the power capacity mix from 31% in 2018 to 18.6% in 2040.

Petronas is partnering with various companies to explore the potential and viability of CCS technology, making Malaysia a key CCS hub in the ASEAN region.

ELECTRICITY

Electricity consumption in Peninsular Malaysia is projected to grow at around 4.5%/year over the medium term, driven by economic expansion, population growth, electrification, rising cooling demand, and accelerating data-centre development.

Announced projects of data-centres now amount to around 3.4 GW of planned capacity (accounting for roughly 60% of ASEAN's development pipeline as of July 2026), and, if fully built out, could consume up to 11% of Malaysia's electricity demand by 2035. In June 2026, the government indicated that electricity consumption from the data centre sector is expected to surge from 7% (equivalent to 10.5 TWh in 2026) to 31%, or 73 TWh by 2035. Peak demand in Peninsular Malaysia is projected to grow at a compound annual growth rate (CAGR) of 5.1% between 2026 and 2035, increasing from 21.3 GW in 2026 to 33.5 GW by 2035.

The country's 2023 National Energy Transition Roadmap (NETR) aims to raise renewable energy capacity from about 13 GW in 2025 to 68 GW by 2050, with renewables making up 70% of total installed capacity.

The NEP 2022-2040 targets 18.4 GW of renewable capacity (including hydro) in 2040 and a reduction in the share of coal in the capacity mix from 31% in 2018 to 18.6% in 2040.As of July 2026, nearly 2.3 GW of new power capacity is currently under construction, including 1.6 GW of hydropower, 1 GW of gas-fired generation, and 0.3 GW of solar. Around 12 GW of additional capacity has been announced, comprising 6.5 GW of solar, 4.6 GW of gas-fired generation, and 1 GW of hydropower.

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EnerFuture: Country-level Energy Forecasts



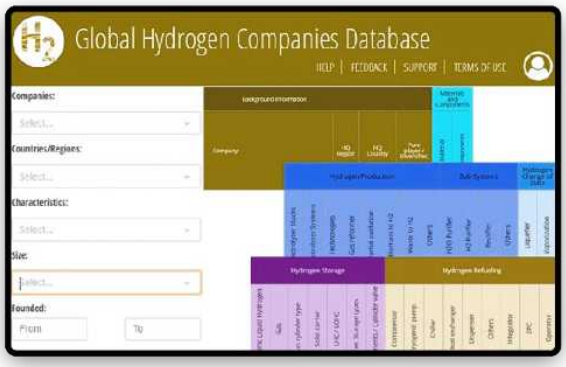
Forecasts through 2050 for demand, prices, CO_2 emissions and power generation. Forecasts for oil, gas, power, coal and renewables by sector for more than 65 individual countries and aggregates.

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